CIE AS PHYSICS (9702) • QUESTION PAPER

10.1 Practical Circuits

Syllabus Section: 10. D.C. Circuits **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

A battery is marked 9.0 V.

What does this mean ?

A each coulomb of charge from the battery supplies 9.0 J of energy to the whole circuit

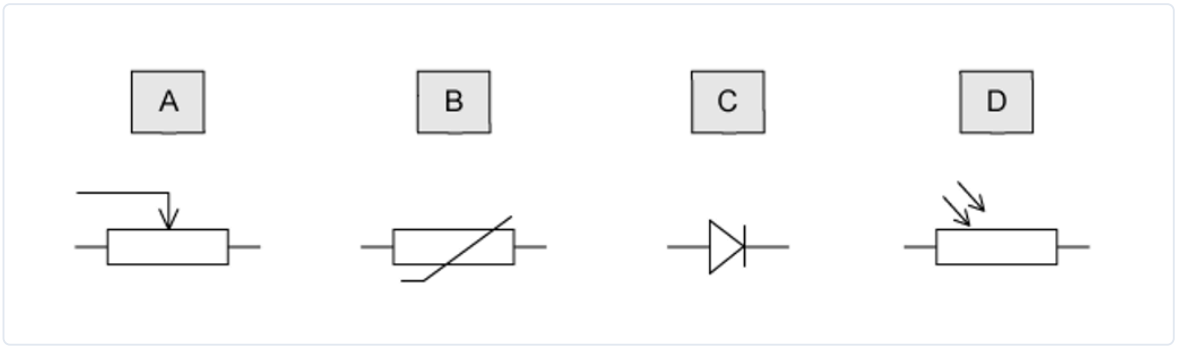
B the battery supplies 9.0 J to an external circuit for each coulomb of charge

C the potential difference across any component connected to the battery will be 9.0 V

D there will always be 9.0 V across the battery terminals

Question 2 Easy [1 mark]

Which symbol represents a component whose resistance is designed to change with temperature?



A

B

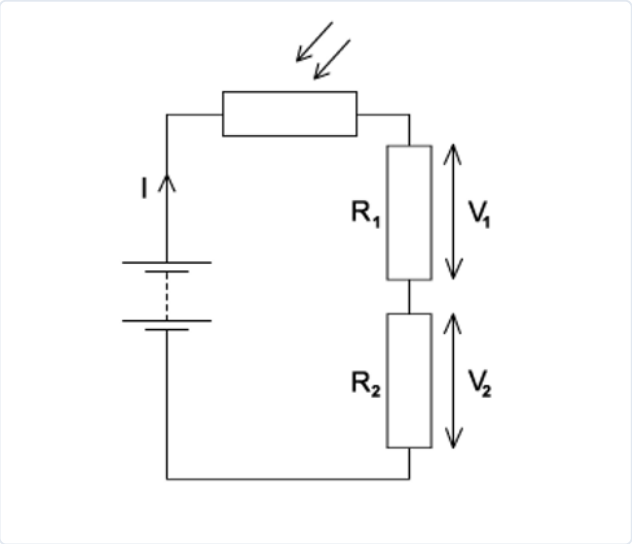
C

D

Question 3 Easy [1 mark]

In the circuit shown, a light-dependent resistor (LDR) is connected to two resistors R_1 and R_2.

The potential difference (p. d.) across R_1 is V_1 and the p .d. across R_2 is V_2. The current in the circuit is I .



Which statement about this circuit is correct ?

A the current I increases when the light intensity decreases

B the LDR is an ohmic conductor

C the p .d. V_2 increases when the light intensity decreases

D the ratio $\frac{V_1}{V_2}$ is independent of light intensity

Question 4 Easy [1 mark]

A cell of e. m .f. E delivers a charge Q to an external circuit.

Which statement is correct?

A the energy dissipation in the external circuit is EQ

B the energy dissipation within the cell is EQ

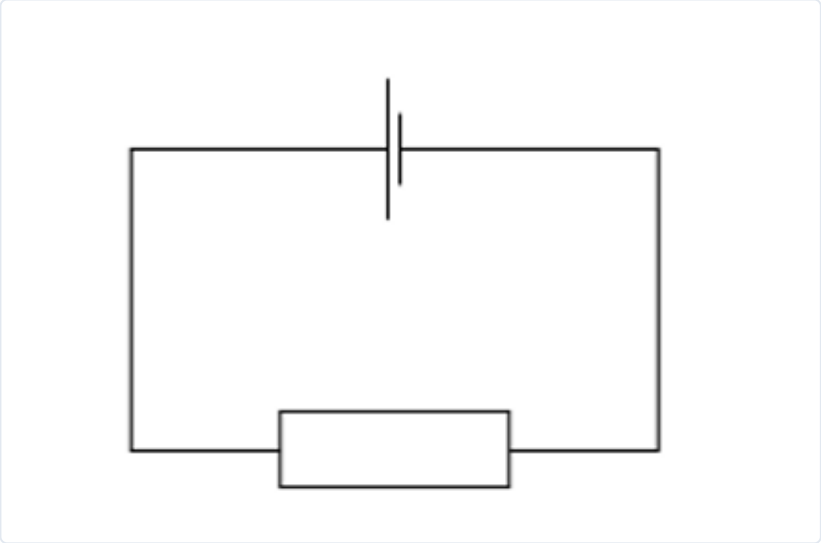
C the external resistance is EQ

D the total energy dissipation in the cell and the external circuit is EQ

Question 5 Easy [1 mark]

A cell is connected to a resistor.

At any given moment, the potential difference across the cell is less than its electromotive force.



Which statement explains this ?

A the cell is continually discharging

B the connecting wire has some resistance

C energy is needed to drive charge through the cell

D power is used when there is a current in the resistor

Question 6 Easy [1 mark]

Four statements about potential difference or electromotive force are listed.

1	It involves the transfer of energy to components
2	It involves the transfer of energy into the circuit
3	It is the energy per unit charge to move charge right round a circuit
4	It is the work done per unit charge by the charge moving from one point to another

Which statements apply to potential difference and which apply to electromotive force?

	Potential difference	**Electromotive force**
A	1 and 3	2 and 4
B	1 and 4	2 and 3
C	2 and 3	1 and 4
D	2 and 4	1 and 3

A

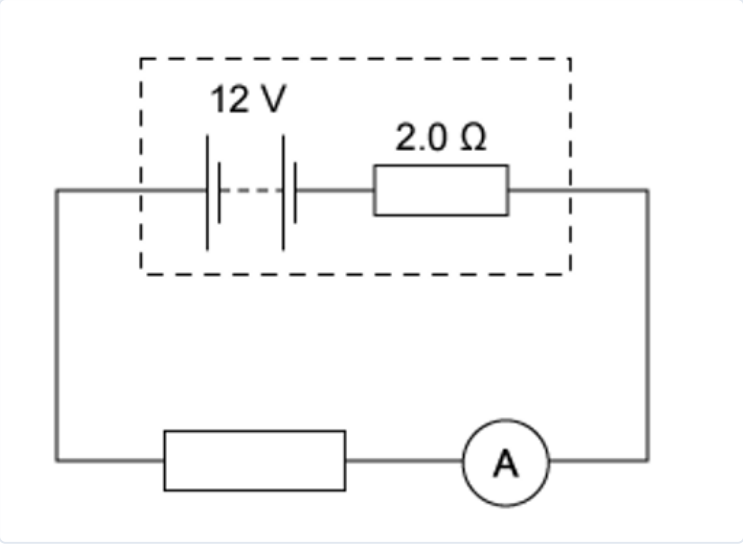
B

C

D

Question 7 Medium [1 mark]

A battery of e .m .f. 12 V and internal resistance 2.0 Ω is connected in series with an ammeter of negligible resistance and an external resistor. External resistors of various different values are used.



Which combination of current and resistor value is **not** correct?

	Current / A	External resistor value / Ω
A	1.0	10.0
B	1.2	8.0
C	1.5	6.0
D	1.8	4.0

A

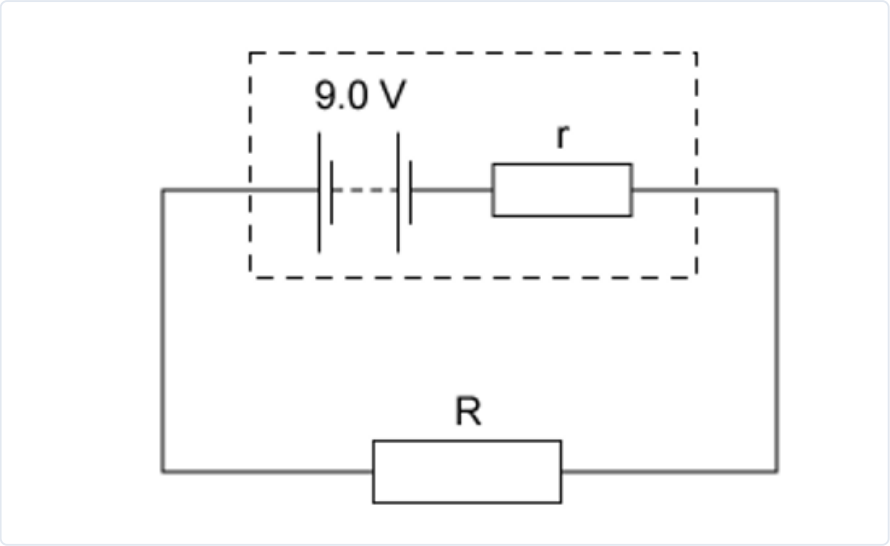
B

C

D

Question 8 Medium [1 mark]

A simple circuit is formed by connecting a resistor of resistance *R* between the terminals of a battery of electromotive force (e .m .f.) 9.0 V and constant internal resistance *r* .



A charge of 6.0 C flows through the resistor in a time of 2.0 minutes, causing it to dissipate 48 J of thermal energy.

What is the internal resistance *r* of the battery ?

A 0.17 Ω

B 0.33 Ω

C 20 Ω

D 160 Ω

Question 9 Medium [1 mark]

A battery and a resistor are in series. The e.m.f of the circuit is 15 V and the resistance of the resistor is 8.0 Ω.

Given that the current through the circuit is 1.5 A, what is the value of the lost volts from the battery?

A 3 V

B 4 V

C 9 V

D 12 V

Question 10 Medium [1 mark]

Which answer option summarises the difference between potential difference and e.m.f?

A Potential difference is the energy transferred into the circuit and e.m.f. is the energy transferred away from the circuit

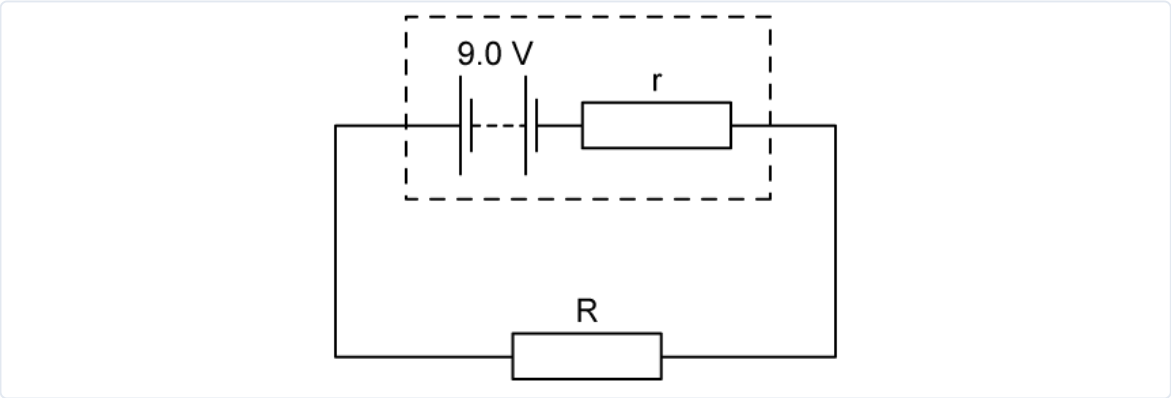
B e.m.f. is the energy transferred into the circuit and potential difference is the energy transferred out of the circuit

C Potential difference and e.m.f. are chemical energy from the cell converted to electrical energy in the components

D Potential difference is the chemical energy of the cell transferred to the electrical energy of the charges and e.m.f. is the electrical energy of the charges converted to other forms of energy in the components

Question 11 Medium [1 mark]

A resistor of resistance R is connected to the terminals of a battery of emf 9.0 V and internal resistance r.



A charge of 240 C flows through the resistor in two minutes. The energy dissipated in the resistor as a result is 1440 J.

What is the internal resistance r of the battery?

A 1.5 Ω

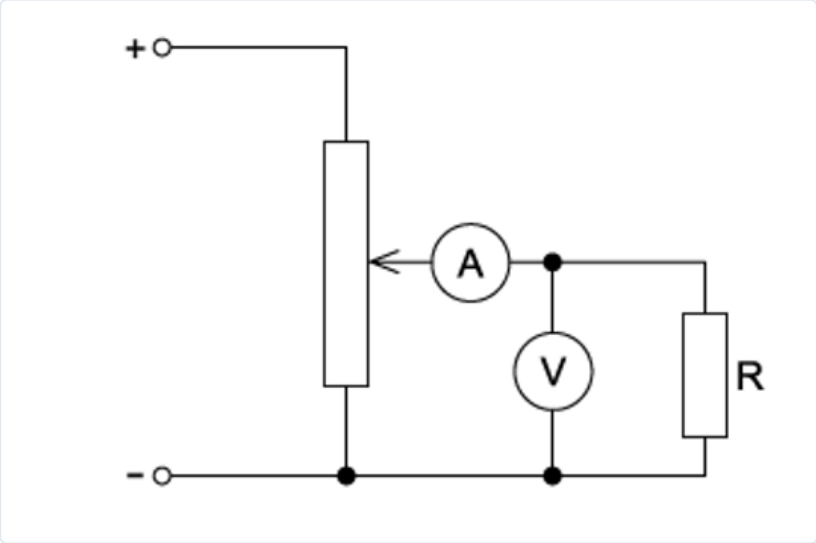
B 2.0 Ω

C 3.0 Ω

D 4.5 Ω

Question 12 Hard [1 mark]

In the circuit below, a voltmeter of resistance R_V and an ammeter of resistance R_A are used to measure the resistance R of the fixed resistor.



Which condition is necessary for an accurate value to be obtained for R ?

A R is much smaller than R_V

B R is much smaller than R_A

C R is much greater than R_V

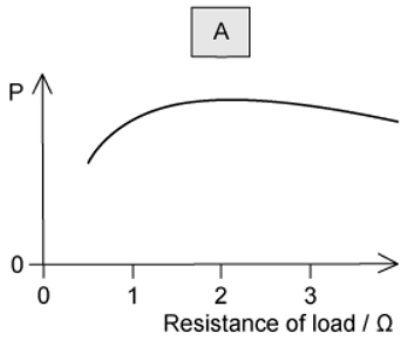
D R is much greater than R_A

Question 13 Hard [1 mark]

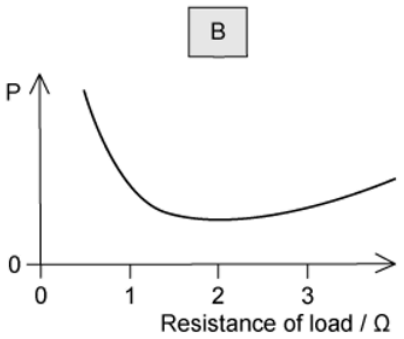
A power supply of electromotive force (e .m .f.) 12 V and internal resistance 2 Ω is connected in series with a load resistor. The value of the load resistor is varied from 0.5 Ω to 4 Ω.

Which graph shows how the power *P* dissipated in the load resistor varies with the resistance of the load resistor ?

A



B



C



D



A

B

C

D

Question 14 Hard [1 mark]

A relay is required to operate 800 m from its power supply. The power supply has negligible internal resistance. The relay requires 16.0 V and a current of 0.60 A to operate.

A cable connects the relay to the power supply, and two of the wires in the cable are used to supply power to the relay.

The resistance of each of these wires is 0.0050 Ω per meter.

What is the minimum output e .m .f. of the power supply ?

A 16.6 V

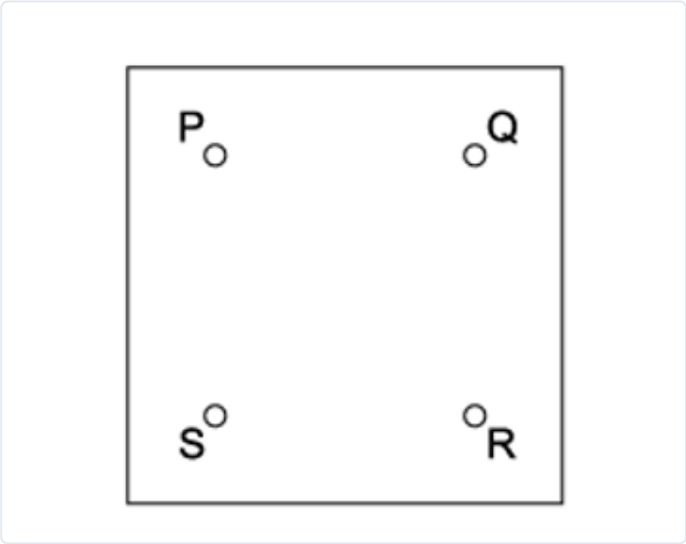
B 18.4 V

C 20.8 V

D 29.3 V

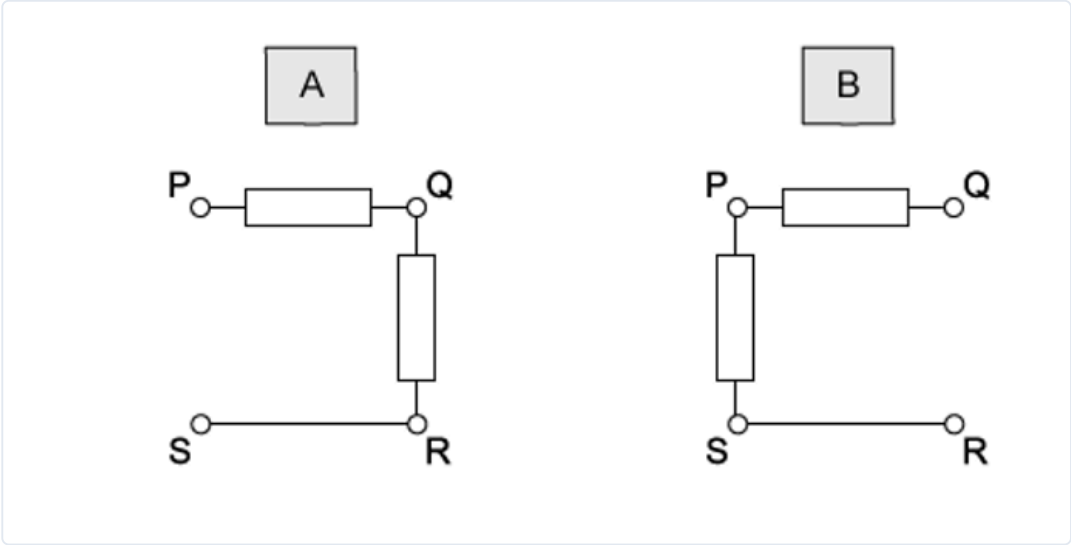
Question 15 Hard [1 mark]

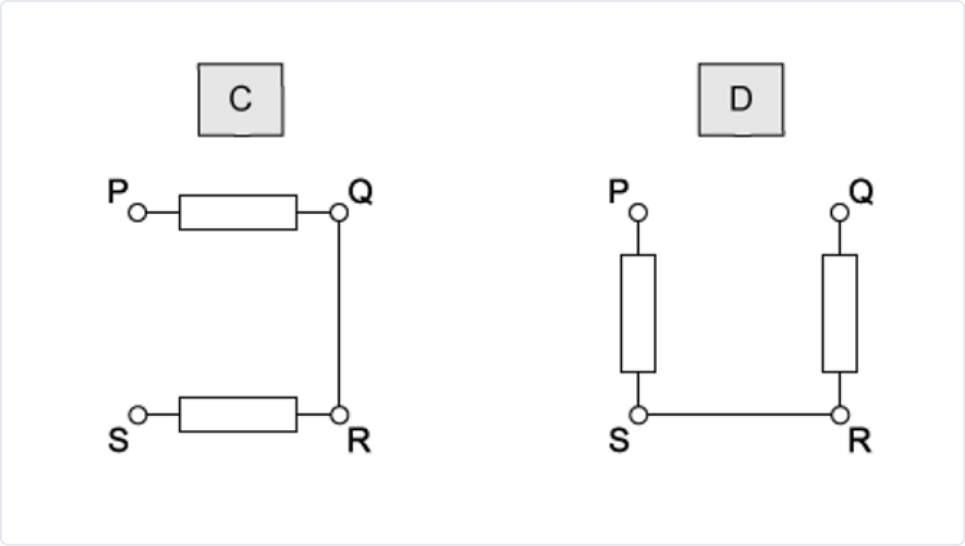
A box with four terminals P, Q, R and S contains two identical resistors.



When a battery of electromotive force (e .m. f.) *E* and negligible internal resistance is connected across PS, a high-resistance voltmeter connected across QR reads $\frac{E}{2}$

Which diagram shows the correct arrangement of the two resistors inside the box ?





A

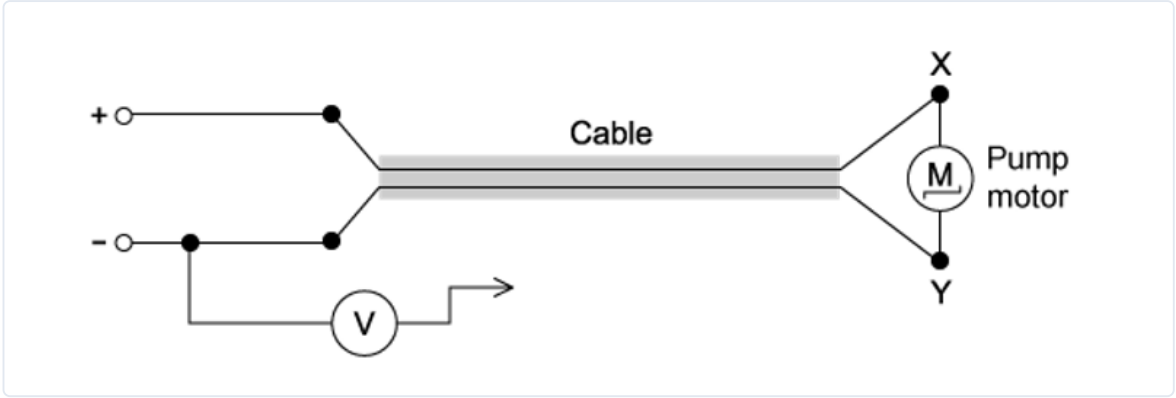
B

C

D

Question 16 Hard [1 mark]

The diagram shows the electric motor for a garden pump connected to a 24 V power supply by an insulated two-core cable.



The motor does not work so, to find the fault, the negative terminal of a voltmeter is connected to the negative terminal of the power supply, and its other end is connected in turn to terminals X and Y at the motor.

Which row represents two readings and a correct conclusion?

	Voltmeter reading when connected to X / V	Voltmeter reading when connected to Y / V	Conclusion
A	24	0	break in positive wire of cable
B	24	12	break in negative wire of cable
C	24	24	break in connection within the motor
D	24	24	break in negative wire of cable

A

B

C

D